

Mitigating Gender Bias in Toxicity Detection via Data Augmentation and Gender Term Masking

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Abstract

Toxicity detection systems are widely deployed to moderate online platforms, yet they often inherit and reinforce social biases. In particular, prior work has shown that comments mentioning identity terms receive inflated toxicity scores even when context is neutral, undermining both fairness and trust in such systems. We investigate two simple but conceptually distinct data augmentation strategies—*gender swapping*, which balances representation by replacing gendered terms with their counterparts, and *gender masking*, which removes explicit gender cues. Both approaches were applied to fine-tune BERT and RoBERTa models on the Jigsaw Unintended Bias in Toxicity Classification dataset, and evaluated with subgroup AUC metrics as well as a dedicated counterfactual fairness test set. Our results show that masking consistently improves fairness without harming accuracy, while swapping yields model-dependent gains: highly effective for BERT, but less impactful for RoBERTa. Counterfactual evaluations reveal disparities more sharply than group metrics, underscoring the value of individual-level tests. Overall, our findings suggest that lightweight augmentations can meaningfully reduce bias, with masking serving as a robust general-purpose strategy and swapping offering complementary benefits in certain architectures. We conclude by highlighting the importance of combining subgroup and counterfactual evaluations to obtain a fuller picture of unintended bias in NLP systems.

Code and Data Availability

The source code and resources for this project are available at: <https://github.com/maysfar/Gender-Bias-in-Toxicity-NLP-project>

1 Introduction

Toxicity detection systems play an important role in maintaining healthy online communities. Yet, these systems often inherit and reinforce societal biases, particularly among marginalized groups. Prior research has shown that toxicity classifiers unfairly assign higher toxicity scores to comments containing identity terms, even when the context is neutral [6]. Such disparities echo broader findings in natural language processing, where models trained on human text frequently encode stereotypical associations between gender and social roles [2, 20]. These biases matter not only from a fairness perspective, but also because they reduce the reliability and trustworthiness of machine learning systems deployed in real-world contexts [1].

Bias in machine learning systems arises when statistical learning objectives exploit spurious correlations present in the



Google's Perspective API

"I am a man" → 20% toxic

"I am a woman" → 41% toxic

Figure 1: "Algorithmic Fairness in Post-Processed Toxicity Text Classification." [12]

data. If gendered words consistently correlate with negative labels, models will internalize and reinforce those associations. This process can occur at multiple levels: during data collection (imbalanced representation of groups), annotation (biased human judgments), representation learning (embedding stereotypes), and modeling choices (loss functions or architectures that amplify skew) [8, 9]. As a result, biased models can produce both; some groups are treated unfairly, and stereotypes are reinforced [4, 18]. Mitigating such bias is therefore both a scientific challenge and a social necessity. Effective interventions must reduce gaps while preserving predictive performance. Prior work has explored approaches ranging from adversarial learning [18] to instance weighting [19], and various augmentation techniques [21]. In this work, however, we focus on two strategies that closely mirror human intuitions about fairness: *gender swapping* and *gender masking*. Swapping replaces gendered terms with their counterparts (e.g., "he" ↔ "she"), reflecting the principle of granting *equal opportunities*. Masking, by contrast, removes gender cues (e.g., replacing with MASK, echoing the idea of striving for *impartiality* by not considering gender when it is relevant. We argue that these two are not only computationally simple and effective, but also conceptually aligned with how humans attempt to reduce bias in decision-making.

In this project, we evaluate these two strategies on the Jigsaw Unintended Bias in Toxicity Classification dataset [3, 11]. Building on prior studies [5, 13, 21], we fine-tune the BERT and RoBERTa models with gender swap and gender masking augmentation and assess their effectiveness using subgroup and bias-sensitive metrics.

Our contributions are threefold. First, we implement and compare gender swap and gender masking in the context of toxicity detection. Second, we analyze their impact across nuanced fairness metrics, highlighting both improvements and

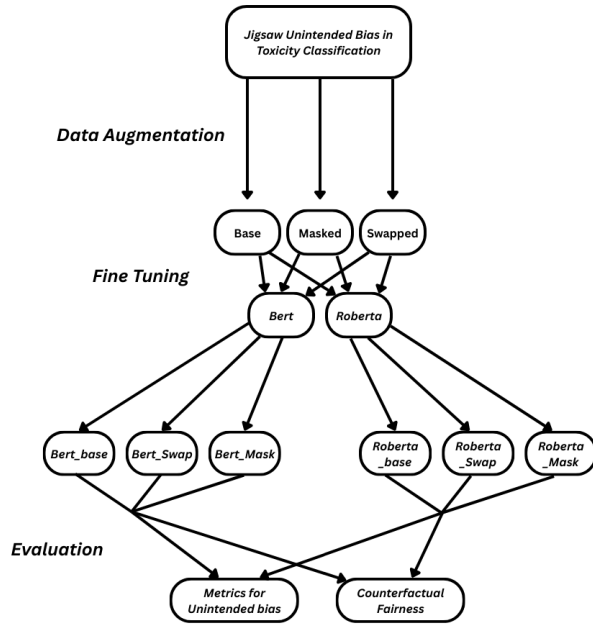


Figure 2: Methodology pipeline

limitations. Finally, we frame these strategies in relation to human notions of fairness *equal opportunity* and *impartiality*, to connect computational interventions with human-inspired reasoning about bias mitigation.

2 Related Work

Bias in NLP. Prior surveys document how social biases appear in datasets, models, and applications [1, 8, 9]. These works highlight that bias arises from multiple sources, including data collection, annotation, and modeling. **Mitigation strategies.** Model-level approaches include adversarial learning [4, 18], while data-level methods include reweighting biased examples [19]. Augmentation methods such as counterfactual rewriting [21] generate balanced examples to reduce bias. **Toxicity detection.** The Jigsaw dataset [3, 11] revealed that toxicity classifiers disproportionately flag comments mentioning identity terms [6]. Work on abusive language detection has explored gender swap [13] and masking [5] as augmentation strategies. **Our focus.** We build on this prior work by systematically comparing swap and masking on the Jigsaw dataset using large pre-trained models (BERT and RoBERTa), and by framing these methods through human fairness principles of *equal opportunity* (swap) and *impartiality* (mask).

3 Methodology

3.1 Overview

Our approach followed a multi-stage pipeline. We first fine-tuned two transformer architectures—RoBERTa-base and BERT-base-uncased—on the Jigsaw Unintended Bias in Toxicity Classification dataset [10], using the official training and test splits.

The baseline models were assessed with standard toxicity classification metrics as well as fairness-oriented measures drawn from the Nuanced Metrics for Measuring Unintended Bias paper [3], and further stress-tested with a counterfactual fairness approach inspired by Garg et al. [7]. To mitigate observed gender bias, we then re-trained both models on augmented versions of the dataset, applying two interventions:

- (1) gender term masking, where gendered words were replaced by a neutral placeholder, and
- (2) gender swapping, where counterfactual variants with flipped gender terms were added to the training data.

These debiased models were evaluated with the same metrics as the baselines, with particular attention to preserving overall toxicity detection performance while reducing subgroup disparities.

3.2 Dataset

We conducted all experiments using the Jigsaw Unintended Bias in Toxicity Classification dataset [10], a large-scale collection of online comments annotated for toxicity and demographic identity mentions. The training split contains approximately 1.78 million comments, each assigned a continuous toxicity score between 0 and 1. Annotators also indicated which identity terms appeared in each comment; for every identity (e.g., male, female), the dataset records the fraction of annotators who marked its presence. For this project, we restricted our analysis to the male and female identity columns, as our research focused on mitigating gender-related bias.

The corresponding test set of ~96,000 comments was used for evaluation under the same preprocessing pipeline.

Preprocessing. We applied minimal text cleaning in order to preserve linguistic cues while removing irrelevant noise. Specifically, we:

- Stripped leading and trailing whitespace and normalized Unicode to NFC form.
- Replaced URLs with the token <URL> and user handles with <USER>.
- Removed line breaks (\n) and similar artifacts.
- Collapsed multiple consecutive whitespace characters into a single space.

This light-touch cleaning ensured consistency across samples while minimizing the risk of discarding patterns useful for toxicity detection.

3.3 Data Augmentation

To investigate whether targeted interventions could mitigate gender bias in toxicity detection, we constructed augmented variants of the Jigsaw dataset using two complementary strategies: gender masking and gender swapping.

3.3.1 Gender Masking. In this approach, proposed by Wadhwa et al. [17], every explicitly gendered token in a comment was replaced with a neutral placeholder <GENDER>. By removing access to direct gender cues, the model was encouraged

to base its toxicity predictions on broader linguistic context rather than identity terms.

3.3.2 Gender Swapping. For each comment containing gendered words, we generated an additional counterfactual comment in which all gendered tokens were replaced with their opposite-gender counterparts. This procedure preserved the original toxicity label while balancing exposure to both male- and female-referenced forms. Comments without gendered terms were left unchanged. After augmentation, the size of the training dataset increased from approximately 1.78 million to 2.4 million comments. This approach was explored in the paper by Park et al. [13].

3.3.3 Gendered Word Dictionary. The lexicon of gendered word pairs used for both masking and swapping was generated with the assistance of ChatGPT. The prompt requested a comprehensive, cleaned dictionary of strict male–female pairs, covering pronouns, family roles, professions, titles, honorifics, insults, slang, anatomy-based terms, and modern internet archetypes. The final dictionary contained about 200 unique pairs, each appearing only once to ensure consistency during augmentation.

3.4 Test Dataset Gendering

A challenge we encountered during evaluation was that the Jigsaw test dataset does not always provide explicit gender information necessary for computing fairness metrics such as Subgroup AUC, BPSN AUC, and BNSP AUC. For these metrics to be meaningful, each comment must be classified as primarily male, female, or neutral.

The dataset includes identity columns that capture the fraction of annotators who marked the presence of specific identities. These were helpful in many cases:

- When both male and female scores were 0, the comment was unambiguously neutral.
- When one score was >0.5 and the other <0.5 , we assigned the gender corresponding to the majority label.

However, ambiguous cases remained. These included comments where:

- Both male and female scores exceeded 0.5, suggesting annotators believed both identities were strongly present.
- Both scores fell between 0 and 0.5, making it unclear which identity (if any) was dominant.

Gender Classification. To resolve these ambiguous cases, we developed a gender classification model trained on an auxiliary dataset of gender-labeled sentences. This dataset consisted of:

- ~8,000 samples from the Dataset for Gendered Language [15].
- ~3,000 additional samples generated using large language models (GPT-5, GPT-4, Claude 3.5, Grok, DeepSeek, and Gemini).

The combined dataset contained 11,061 labeled samples, with each sentence annotated as male or female depending on the most prevalent identity reference.

We fine-tuned two transformer models on this dataset:

- RoBERTa-base (achieving 98.1% accuracy)
- DeBERTa v3-base (achieving 98.5% accuracy)

Given its superior performance, we used the fine-tuned DeBERTa v3-base classifier to automatically label ambiguous test comments as male or female, thereby enabling robust and consistent fairness evaluation.

3.5 Finetuning RoBERTa and BERT

To evaluate both baseline and bias-mitigation strategies, we fine-tuned two transformer models—RoBERTa-base and BERT-base-uncased—on three different versions of the Jigsaw dataset:

- Original dataset (baseline condition)
- Gender-masked dataset (bias mitigation via masking)
- Gender-swapped dataset (bias mitigation via counterfactual augmentation)

This design yielded a total of six fine-tuned models, enabling direct comparisons across architectures and augmentation strategies.

3.5.1 Data Splits and Tokenization. We partitioned the Jigsaw training data into 80% training and 20% validation splits. All comments were tokenized with a maximum sequence length of 256 tokens, applying truncation to longer texts and padding to shorter ones.

3.5.2 Task Framing. Because the Jigsaw labels are continuous scores between 0 and 1, we formulated toxicity detection as a regression problem rather than binary classification. Each model produced a single scalar prediction per comment, optimized to approximate the toxicity score.

3.5.3 Training Configuration. All models were trained using the HuggingFace Trainer API, under the following configuration:

- Optimizer & Learning Rate: AdamW with a learning rate of 3×10^{-5} .
- Batch Size: 128 for both training and evaluation.
- Epochs: 2 full passes through the training data.
- Weight Decay: 0.01 for regularization.
- Precision: Mixed precision (bfloat16) to improve efficiency on A100 GPUs.
- Checkpointing & Evaluation: Checkpoints were saved at the end of each epoch, with the best-performing model selected based on validation performance. Evaluation was conducted once per epoch.

3.5.4 Metrics and Monitoring. During training, we monitored mean squared error (MSE), mean absolute error (MAE), and R^2 to ensure models learned meaningful mappings to toxicity scores. After training, fairness-oriented metrics and counterfactual fairness evaluations were computed on the held-out Jigsaw test set.

4 Evaluation

4.1 Bias Evaluation Metrics

This section formalizes a set of fairness metrics used to evaluate our toxicity classifier on gender-specific subgroups (male and female). Following the framework proposed by Borkan and colleagues [3], we consider a binary classification task where each example i has a ground-truth label $y_i \in \{0, 1\}$ (toxic and non-toxic), a model score $\hat{y}_i \in [0, 1]$ that reflects the model’s confidence in the “toxic” class, and a Boolean indicator $g_i \in \{0, 1\}$ specifying whether the text belongs to the gender subgroup currently under evaluation.

Throughout this section we write

- $\mathcal{D}^+$ and $\mathcal{D}^-$ for the sets of positive and negative examples in the background (that is, $\{i \mid y_i = 1, g_i = 0\}$ and $\{i \mid y_i = 0, g_i = 0\}$, respectively),
- $\mathcal{D}_g^+$ and $\mathcal{D}_g^-$ for the sets of positive and negative examples in the gender subgroup (that is, $\{i \mid y_i = 1, g_i = 1\}$ and $\{i \mid y_i = 0, g_i = 1\}$). In practice we compute each metric twice, treating male comments as the subgroup ($g_i = 1$) and female comments as the subgroup in turn.

All of the metrics below are *threshold-agnostic*: they summarize the entire score distribution rather than the performance at a fixed decision threshold. This property is desirable when models are deployed with different thresholds across applications. We use the area under the receiver operating characteristic (ROC-AUC). For any classifier, AUC measures the probability that a randomly chosen negative example will receive a lower score than a randomly chosen positive example, i.e. that the two will be correctly ordered. An AUC of 1.0 means that all negative/positive pairs are correctly ordered, with all negative items receiving lower scores than all positive items.

4.1.1 AUC-based fairness metrics. The first three metrics evaluate separability between different parts of the data. we define:

Subgroup AUC. The *subgroup AUC* measures separability within the subgroup:

$$\text{AUC}(\mathcal{D}_g^- + \mathcal{D}_g^+).$$

A high value indicates that toxic and non-toxic comments within the subgroup are well separated. A low value indicates that the model does not distinguish well between toxic and non-toxic comments mentioning the subgroup.

BPSN AUC. The *background positive, subgroup negative (BPSN) AUC* compares positive examples from the background with negative examples from the subgroup:

$$\text{AUC}(\mathcal{D}^+ + \mathcal{D}_g^-).$$

Low BPSN AUC indicates that non-toxic comments mentioning the subgroup often receive disproportionately high scores, causing them to outrank positive background examples.

BNSP AUC. The *background negative, subgroup positive (BNSP) AUC* compares negative background examples with positive subgroup examples:

$$\text{AUC}(\mathcal{D}^- + \mathcal{D}_g^+).$$

Low BNSP AUC means that positive subgroup examples receive lower scores than negative background examples, leading to a high risk of false negatives for the subgroup.

These AUC-based metrics are inherently robust to variations in group prevalence because each metric compares score distributions on disjoint subsets of examples. By ensuring that all positives (or all negatives) in each AUC come from a single group, mis-orderings involving that group cannot be diluted by other data.

4.1.2 Average Equality Gap metrics. The AUC metrics above primarily detect large mis-orderings across classes. To capture more subtle “score shifts” where subgroup and background distributions have different means but remain separable, Borkan et al. introduced the *Average Equality Gap (AEG)* metrics. These metrics are derived from the equality gap (difference in true positive rates at a fixed threshold) by averaging over all thresholds. They also rely on the fact that the concordance index used in survival analysis reduces to the Wilcoxon–Mann–Whitney statistic when applied to static binary discrimination, as noted by Park and colleagues [14]. The same equivalence is observed in recent work on AUC optimization by Sun et al. [16]. In particular, if X and Y are two collections of scores with $|X| = n_X$ and $|Y| = n_Y$, then the AUC can be written in terms of the Mann–Whitney U statistic, denoted $\text{MWU}(X, Y)$, as

$$\text{AUC}(X + Y) = \frac{\text{MWU}(X, Y)}{n_X n_Y}. \quad (1)$$

Using this relationship, the AEG metrics can be expressed concisely as follows:

Positive AEG..

$$\text{AEG}^+ = \frac{1}{2} - \frac{\text{MWU}(\mathcal{D}_g^+, \mathcal{D}^+)}{|\mathcal{D}_g^+| |\mathcal{D}^+|}.$$

Equivalently, $\text{AEG}^+ = \frac{1}{2} - \text{AUC}(\mathcal{D}_g^+ + \mathcal{D}^+)$. Intuitively, it is one half minus the probability that a toxic subgroup example receives a higher score than a toxic background example. Positive values mean the subgroup’s toxic scores tend to be lower than those of the background (a leftward shift), whereas negative values mean the scores are higher.

Negative AEG..

$$\text{AEG}^- = \frac{1}{2} - \frac{\text{MWU}(\mathcal{D}_g^-, \mathcal{D}^-)}{|\mathcal{D}_g^-| |\mathcal{D}^-|}.$$

Again $\text{AEG}^- = \frac{1}{2} - \text{AUC}(\mathcal{D}_g^- + \mathcal{D}^-)$. A positive value indicates that non-toxic subgroup examples receive higher scores on average, increasing the risk of false positives, while a negative value indicates lower scores for the subgroup’s non-toxic examples.

In synthetic experiments, Borkan and colleagues [3] showed that small upward or downward shifts in subgroup scores leave the AUC-based metrics unchanged but produce non-zero

AEG values. Larger shifts are reflected in both the AEGs and the BPSN/BNP AUCs, so considering both types of metrics provides a nuanced view of unintended bias.

4.2 Counterfactual Fairness

To complement group-level fairness metrics, we also evaluated model bias at the individual level using the Counterfactual Fairness Gap (CFG) — a metric that measures how often a model’s prediction would change if one intervened on a protected attribute (e.g., gender) while holding everything else constant.

In the context of text classification, this aligns with notions such as counterfactual token fairness (CTF), where small identity-based perturbations in a sentence (e.g. “he → she”) should not systematically change the model’s output unless the semantics genuinely differ.

4.2.1 Counterfactual Test Dataset Construction. We constructed a dedicated counterfactual evaluation dataset with about 500 sentences (i.e. 250 counterfactual pairs) using DeepSeek with the following prompt:

Generate a list of 250 sentence pairs to test gender bias in toxicity detection. Each pair must be identical except for gendered terms (e.g. he ↔ she, his ↔ her, man ↔ woman, boy ↔ girl).

- Produce ≥ 125 pairs of non-toxic sentences (e.g. “He is a doctor.” / “She is a doctor.”)
- Produce ≥ 125 pairs of toxic sentences (e.g. “I hate his attitude.” / “I hate her attitude.”)
- Keep sentences short (max 10 words).
- Cover diverse contexts (jobs, insults, compliments, daily life).
- Output as a list of tuples: (sentence, label (1 if toxic, 0 otherwise), gender).

Each pair thus differs only in gendered terms, while preserving the intended toxicity label.

4.2.2 Measuring Counterfactual Fairness Gap. We then ran each fine-tuned model (from the baseline and augmented variants) on both sentences in each pair, and computed the prediction gap per pair (i.e. the absolute difference in predicted toxicity score). From these pairwise gaps, we derived summary statistics:

- **num_pairs**: total number of evaluated pairs
- **mean_gap**: mean absolute difference across all pairs
- **median_gap**: median of the absolute differences
- **max_gap**: maximum observed difference
- **non_toxic_mean_gap**: mean gap over pairs labeled non-toxic
- **toxic_mean_gap**: mean gap over pairs labeled toxic

By comparing these gaps across models, we gauge how robust each variant is to gender perturbations at the individual level, and whether bias mitigation reduces unexpected sensitivity to identity changes.

5 Results

5.1 Overall Accuracy

Across both BERT and RoBERTa, models achieved consistently high toxicity-prediction accuracy, clustered around 95–96%. Fine-tuning on augmented datasets—via gender masking or gender swapping—did not degrade performance, as reported in Table 1.

For **BERT**, the baseline reached an accuracy of 0.9571. The masked variant achieved 0.9570, and the swapped variant reached 0.9561, showing virtually no change.

For **RoBERTa**, the baseline accuracy of 0.9570 was preserved under masking, which yielded 0.9571, and even slightly improved under swapping with an accuracy of 0.9574. These results indicate that fairness interventions can be applied without sacrificing predictive reliability.

Model	Accuracy
RoBERTa	
tc-roberta	0.957013
tc-roberta_mask	0.957075
tc-roberta_swap	0.957364
BERT	
tc-bert	0.957065
tc-bert_mask	0.956972
tc-bert_swap	0.956084

Table 1: Overall accuracy for RoBERTa and BERT variants

5.2 Subgroup Fairness

Despite strong overall accuracy, subgroup analyses revealed measurable disparities between male and female identity groups, as shown in Table 2.

For **BERT-base**, the subgroup AUC was 0.939 for male and 0.944 for female comments, with Pos. AEG values of 0.053 (male) and 0.081 (female), and Neg. AEG values of -0.198 (male) and -0.194 (female). This indicates a slight advantage for female positives and a stronger disadvantage for male negatives. After applying data augmentation, the gaps narrow. In **BERT-mask**, the female AUC rises to 0.946, while male AUC remains comparable at 0.936. More importantly, Neg. AEG for females shifts from -0.194 to -0.204, while for males it stays near -0.200, reducing the imbalance in how male and female negatives are penalized. In **BERT-swap**, the AUCs remain balanced (0.934 male vs. 0.943 female), and Neg. AEG becomes less asymmetric (-0.195 male vs. -0.178 female), suggesting that swapping reduces the earlier disparity where male negatives were disproportionately disadvantaged.

For **RoBERTa-base**, the subgroup AUC was 0.937 for male and 0.944 for female comments. Bias appears in the AEG metrics: male Pos. AEG is 0.053 versus 0.071 for female, while both subgroups show the same Neg. AEG of -0.192. In the augmented models, these gaps shrink. **RoBERTa-mask** increases female AUC slightly to 0.946 while maintaining male AUC at 0.934, and it also raises Pos. AEG for both groups (0.068 male, 0.087 female) without widening the male–female gap in Neg. AEG (male -0.195, female -0.193). **RoBERTa-swap** further stabilizes results: AUCs stay high (0.936 male, 0.946 female), Pos. AEG differences remain modest (0.052 vs. 0.074), and Neg. AEG values are nearly identical (-0.194 vs. -0.195), indicating that the penalization of subgroup negatives is now more evenly distributed.

5.3 Counterfactual Fairness

The clearest evidence of bias—and mitigation—appears in counterfactual tests, which measure how predictions shift when gendered terms are swapped. These results are presented in Table 3. At baseline, **BERT** showed a mean gap of 0.0327, with toxic sentences particularly affected, showing a gap of 0.0602. **RoBERTa** performed somewhat better, with a mean gap of 0.0228 and a toxic gap of 0.0444. Interventions substantially reduced these disparities. In **BERT**, masking halved the mean gap to 0.0170, and swapping reduced it further to 0.0064, representing approximately an 80% drop compared to baseline. In **RoBERTa**, masking lowered the mean gap from 0.0228 to 0.0153, while swapping left it nearly unchanged at 0.0225. This suggests lower initial sensitivity to gender perturbations and less headroom for improvement.

5.4 Summary of Findings

- **Bias is measurable:** Baseline models show gender gaps, with female positives ranked more favorably (higher Pos. AEG) and male negatives more penalized (lower Neg. AEG). Counterfactual tests confirm sensitivity to gender terms.
- **Augmentation reduces disparities:** Masking and swapping narrow AEG gaps. In **BERT**, swapping lessens male–female Neg. AEG imbalance, while masking improves female AUC. In **RoBERTa**, masking boosts subgroup AUC and swapping equalizes Neg. AEG across genders.
- **Accuracy is preserved:** Both interventions retain baseline accuracy (Table 1).
- **Masking is reliable, swapping model-dependent:** Masking improves fairness across settings, while swapping is especially effective for counterfactual fairness in **BERT** and stabilizes subgroup fairness in **RoBERTa**.

Overall, masking is a robust general-purpose strategy, while swapping provides targeted gains—most notably in **BERT**.

6 Discussion

Our results highlight several important insights into how augmentation strategies interact with model architecture and evaluation methodology. First, gender masking consistently

produced fairness improvements across both subgroup and counterfactual evaluations. This can be explained by the fact that masking removes explicit gender tokens altogether, preventing the model from exploiting them as spurious cues. In the Jigsaw dataset, where the presence of identity words often correlates with toxicity labels, eliminating these tokens forces the model to rely on surrounding context. This leads to more stable and generalizable predictions.

By contrast, gender swapping yielded mixed outcomes. While it balances exposure to male- and female-referenced forms, the gender tokens remain visible, leaving open the possibility that the model will continue to anchor predictions on them. The impact of swapping also varied by architecture: in **BERT**, swapping produced strong gains in counterfactual fairness, likely because the additional data acted as a form of regularization for a smaller-capacity model. In **RoBERTa**, however, the benefits were limited, suggesting that its stronger contextual representations already reduced reliance on identity tokens and that simple duplication added less new information.

The differences between evaluation metrics further illustrate why counterfactual testing is critical. Subgroup AUC and related measures summarize performance across diverse groups, including many identities not directly targeted in our augmentations. As a result, gender-specific improvements may appear muted or inconsistent. Counterfactual evaluation, in contrast, provides a direct, pairwise probe of bias: each sentence is compared only to its gender-flipped counterpart. Because our augmentations explicitly targeted these identities, the counterfactual metric captured improvements more sharply, revealing disparities that subgroup metrics only partially reflected.

Taken together, these findings suggest that masking is the more reliable, general-purpose strategy for debiasing toxicity detection, while swapping can provide model-dependent benefits, particularly for architectures with less representational capacity. More broadly, our analysis underscores the importance of using complementary evaluation metrics: group-level scores reveal broad disparities, but counterfactual probes expose fine-grained sensitivities that may otherwise remain hidden.

7 Conclusion

Our study evaluated two augmentation-based strategies for mitigating gender bias in toxicity detection: gender swapping and gender masking. We found that *masking* consistently improved fairness metrics, reducing the reliance of models on gendered terms. *Swapping* also contributed to more balanced performance across groups, though its effects were more mixed compared to masking. Importantly, both approaches preserved predictive accuracy, suggesting that simple data augmentations can effectively enhance fairness without sacrificing task performance.

Model	Group	Subg. AUC	BPSN	BNSP	Pos. AEG	Neg. AEG
RoBERTa						
tc-roberta	Male	0.936726	0.940487	0.973209	0.053112	-0.191928
	Female	0.944461	0.956570	0.966994	0.071148	-0.192203
tc-roberta_mask	Male	0.934146	0.939877	0.971892	0.067585	-0.195022
	Female	0.945551	0.956983	0.966571	0.087254	-0.192941
tc-roberta_swap	Male	0.936485	0.940362	0.973249	0.051723	-0.193878
	Female	0.945576	0.957114	0.967659	0.073627	-0.195182
BERT						
tc-bert	Male	0.939430	0.939616	0.974970	0.053232	-0.197696
	Female	0.944094	0.956000	0.966763	0.080603	-0.193616
tc-bert_mask	Male	0.935670	0.938573	0.973361	0.063373	-0.199831
	Female	0.945649	0.952827	0.968640	0.078918	-0.203798
tc-bert_swap	Male	0.934219	0.934999	0.973555	0.052218	-0.195025
	Female	0.943190	0.955652	0.965035	0.082606	-0.177704

Table 2: Subgroup fairness metrics by gender for RoBERTa and BERT

Model	Mean Gap	Median Gap	Max Gap	Non-toxic Mean Gap	Toxic Mean Gap
RoBERTa					
tc-roberta	0.0228	0.0037	0.2188	0.0012	0.0444
tc-roberta_mask	0.0153	0.0032	0.1982	0.0012	0.0295
tc-roberta_swap	0.0225	0.0045	0.2377	0.0020	0.0431
BERT					
tc-bert	0.0327	0.0095	0.2649	0.0052	0.0602
tc-bert_mask	0.0170	0.0062	0.1650	0.0052	0.0289
tc-bert_swap	0.0064	0.0011	0.1068	0.0007	0.0121

Table 3: Counterfactual fairness metrics for RoBERTa and BERT

8 Limitations and Future Work

This work has several limitations. First, we only fine-tuned pre-trained transformer models (BERT and RoBERTa), rather than training from scratch. Training models from the ground up could provide clearer insight into how augmentations shape learned representations. Second, our interventions targeted only binary gender terms, overlooking broader identity dimensions and intersectional biases. Third, our augmentations were rule-based, which may fail to capture subtler or context-dependent uses of gendered language. Finally, our evaluation was restricted to the Jigsaw dataset and fairness metrics that, while useful, cannot fully capture individual-level fairness.

Future work can extend these directions in multiple ways. Training smaller models from scratch on augmented versus non-augmented corpora would isolate the direct effects of augmentation. Augmentations could also be expanded to other identity attributes such as race, religion, or age, and evaluated for intersectional fairness. More advanced augmentation methods—such as context-aware paraphrasing or LLM-based counterfactual rewriting—may provide more natural training data. Beyond augmentation, integrating swap and mask strategies with contrastive learning could further encourage fairness in representation learning. Cross-lingual and cross-domain studies would test the robustness of these methods in settings where gender is more deeply embedded in grammar.

Resources

Models

- moatazhamza194/tc-bert
- moatazhamza194/tc-roberta
- moatazhamza194/tc-bert_mask
- moatazhamza194/tc-roberta_mask
- moatazhamza194/tc-bert_swap
- moatazhamza194/tc-roberta_swap
- moatazhamza194/gender_classification-deberta

Datasets

- moatazhamza194/gb_train
- moatazhamza194/gb_test
- moatazhamza194/gb_train_mask
- moatazhamza194/gb_train_swap
- moatazhamza194/gb_test_gendered

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